Sungmin Yun

Computer Architectures and Systems for AI

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About Me

I am a **Senior Deep Learning Performance Architect at NVIDIA**, focusing on **AI acceleration and performance optimization**. I have extensive hands-on experience in **accelerator design, hardware profiling, performance modeling, and kernel programming**, with strong expertise in **modern AI architectures** (e.g., MoE, MLA, RAG, diffusion). My research has focused on **maximizing hardware utilization** for LLM serving and accelerating **Multi-LoRA inference** and **RAG pipelines**. I also developed an **open-source end-to-end LLM performance simulator**, LLMSimulator, enabling rapid evaluation of emerging hardware and LLM architectures. My work aims to **maximize the performance and efficiency of AI systems** through hardware–software co-design.

Selected Publications

[MICRO 2024] Duplex: A Device for Large Language Models with Mixture of Experts, Grouped Query Attention, and Continuous Batching, **S. Yun**, K. Kyung, J. Cho, J. Choi, J. Kim, B. Kim, S. Lee, K. Sohn, J. H. Ahn, (Acceptance Rate: **22**% (113 among 497))

[ICS 2024] CLAY: CXL-based Scalable NDP Architecture Accelerating Embedding Layers, **S. Yun**, H. Nam, K. Kyung, J. Park, B. Kim, Y. Kwon, E. Lee, J. H. Ahn, (Acceptance Rate: **36**% (45 among 125))

[ICS 2026] Clone: A Collaborative Multi-device System for Retrieval-Augmented Generation over CXL, S. Ko, W. Doh, E. Na, H. Shim, **S. Yun**, J. So, Y. Kwon, S. Park, S. Roh, M. Yoon, T. Song, E. Lee, J. H. Ahn (**Co-corresponding Author**)

Experience

Senior Deep Learning Performance Architect, NVIDIA — Sep. 2026 – Present
Deep learning application performance analysis

Postdoctoral Researcher, Seoul National University — Sep. 2025 – Sep. 2026
Lead the following research projects (2 manuscripts under review):
- **LLM Serving Systems** – develop system-level guidelines and optimizations for LLMs (e.g., MLA, MoE)
- **Multi-LoRA Inference at Scale** – lead end-to-end accelerator design and implementation
- **RAG Pipeline Acceleration** – co-design search algorithms and hardware for large-scale RAG pipelines

Internship, Samsung Electronics — Mar. 2023 – Aug. 2023
- Designed and implemented a hardware controller in RTL for general matrix–vector multiplication (GEMV) using HBM-PIM
- Analyzed architectural limitations of HBM-PIM and proposed Duplex, an LLM accelerator integrating custom HBMs with xPUs (later published at MICRO 2024)

Education

Seoul National University, Ph.D. in Artificial Intelligence — GPA: 4.1/4.3 — Aug 2020 – Aug 2025

Yonsei University, B.S. in Integrated Information Technology — GPA: 3.6/4.3 — Mar 2017 – Feb 2020

Technical Skills

Programming languages	C/C++, CUDA, Python, System Verilog
ML/LLM Frameworks	Pytorch, vLLM, SGLang, TensorRT-LLM
Profiling, Analysis, Simulators	Nsight Compute, Nsight Systems, VTune, Ramulator

Honors & Awards

Best Ph.D. Dissertation Award Aug 2025
Seoul National University

Best of CAL Award Feb 2023
29th IEEE International Symposium on High-Performance Computer Architecture

Invited Presentations

Poster presentation at Samsung AI Forum Nov 2023

Oral presentation on Best of CAL session at IEEE International Symposium on High-Performance Computer Architecture Feb 2023

Program Committee and Paper Review

Lightweight Program Committee Member, ISCA 2026

IEEE Transactions on Computers

IEEE Transactions on Circuits and Systems for Artificial Intelligence

IEEE Computer Architecture Letters

Future Generation Computer Systems

Additional Publications

[IEEE CAL 2025] SSD Offloading for LLM Mixture-of-Experts Weights Considered Harmful in Energy Efficiency, K. Kyung, **S. Yun**, J. H. Ahn

[IEEE CAL 2025] COSMOS: A CXL-Based Full In-Memory System for Approximate Nearest Neighbor Search, S. Ko, H. Shim, W. Doh, **S. Yun**, J. So, Y. Kwon, S. Park, S. Roh, M. Yoon, T. Song, J. H. Ahn

[HPCA 2025] Anaheim: Architecture and Algorithms for Processing Fully Homomorphic Encryption in Memory, J. Kim, **S. Yun**, H. Ji, W. Choi, S. Kim, J. H. Ahn

[IEEE TC 2023] GraNDe: Efficient Near-Data Processing Architecture for Graph Neural Networks, **S. Yun**, H. Nam, J. Park, B. Kim, J. H. Ahn, E. Lee

[IEEE CAL 2022] GraNDe: Near-Data Processing Architecture With Adaptive Matrix Mapping for Graph Convolutional Networks, **S. Yun**, B. Kim, J. Park, H. Nam, J. H. Ahn, E. Lee, (**Best of CAL Award**)

[MICRO 2021] TRiM: Enhancing Processor-Memory Interfaces with Scalable Tensor Reduction in Memory, J. Park, B. Kim, **S. Yun**, E. Lee, M. Rhu, J. H. Ahn

Patents

• Heterogeneous Computing Memory Device and Artificial Intelligence Computing System Including the Same	US KR (1020250050877)
• Memory Device Capable of Dynamically Switching Between Error Correction Code and Error Detection Code in On-Die Error Correction Code	KR Granted (10-279756) PCT (PCT/KR2021/019871)
• Memory Module and Computing System Including the Same	US (19/017,235) KR (1020240060980) CN (202510116255.2)
• Memory Controller and Operation Method Thereof	US (19/002,383) KR (1020240050775) CN (202411903289.8)
• Method and Apparatus for Memory Device Including Processing-in-Memory (PIM)	US KR (1020250026131)
• Memory System and Operating Method Thereof	KR (1020250195434)